

Rooftop Solar Potential

A city-scale pipeline that scores every rooftop for solar suitability — the engineering evolution of a Grasshopper + Ladybug prototype into a reproducible data pipeline over an entire city.

Project plan · v1 Washington, DC → v2 India

Grows from **SIH 2024 BIPV prototype** (Team Operation_VIJAY · PS SIH1739) — companion deck:

[SIH2024_BIPV_Solar_Prototype_Deck.pdf](#)

Aug 2026 · stack-neutral core, Esri deployment track in the appendix

WHERE THIS COMES FROM

This is **not** a from-scratch portfolio piece. In 2024 I built a working **Grasshopper / Rhino + Ladybug** prototype (Smart India Hackathon, PS SIH1739) that pulled OpenStreetMap footprints, extruded them into a **LOD-1 city model** of central New Delhi, ran **full-year solar-radiation and inter-building shadow simulation**, and computed BIPV potential on both rooftops and facades.

This project scales that instinct — from one neighbourhood in a design tool to an *entire city in a reproducible pipeline*, trading hand-built LOD-1 blocks for LiDAR-derived per-roof geometry and deep-learning roof segmentation. The prototype already proved the hard part (mutual shading between buildings, §7); this version earns the resolution and the scale.

01 The one-line pitch

Score the rooftop-solar suitability of every building in a city — per-roof geometry, orientation, tilt, shading, and irradiance → estimated PV capacity, annual energy, and CO₂ offset — then aggregate to neighbourhood choropleths with an equity lens.

Built entirely from **public data and APIs**, modelled in **open-source Python**, and delivered as an **interactive web map**. The Esri stack is one supported deployment target, not a dependency — see the appendix.

02 Why this project — the bridge

This sits exactly where my history converges, which is the entire point: it is the argument for why I belong in geospatial ML, not a generic ML portfolio piece bolted onto an architecture CV.

- **Architecture + OCAD + Ladybug** → reading a roof as geometry (planes, orientation, tilt, self-shading) and reasoning about solar exposure on built form is *native* to me. I already prototyped this logic in Grasshopper/Rhino; this project takes it from one building in a design tool to a whole city in a pipeline.
- **Solar / SLnko** → I work in solar energy. Rooftop suitability and its sibling, utility-scale siting, are the two core spatial questions of the industry. This is not a hobby topic — it is my domain, made spatial.
- **ML** → the roof-plane / obstruction extraction and the suitability model are genuine machine-learning problems (segmentation, learned ranking), not a GIS overlay dressed up as ML.

03 The core design insight: building resolution × city extent

THE SCALE MOVE

A single building is too small to read as "geospatial." The fix is not to abandon building-level detail — it is to **keep building-level resolution while operating at city-wide extent**, then aggregate up.

That is precisely what **Google Project Sunroof** and **NREL/NLR** rooftop-potential studies do: score tens of thousands of individual roofs across a metro, then roll up to neighbourhood / census-tract choropleths for planning and equity analysis. You get both the per-roof craft *and* the real urban-scale geospatial engineering — large data volumes, tiling, spatial joins, aggregation, spatial statistics.

04 What you'll build — system overview



05 Two modelling modes behind one interface

The geometry front-end is swappable; the rest of the pipeline (radiation → yield → aggregation) is identical downstream. This is what makes the same codebase serve a data-rich US city and a sparse-data Indian one — and it is a direct descendant of the prototype.

High-resolution path — LiDAR per-roof V1 DEFAULT

Clip the LiDAR point cloud per footprint, fit individual roof planes (RANSAC baseline or the ML route in §9), and resolve genuine per-roof slope, aspect, and obstructions. This is the Washington-DC v1 default and the source of the project's resolution story.

Block-model path — LOD-1 extruded footprints HERITAGE + FALLBACK

Extrude footprints to a single height per building (the **LOD-1 model my SIH prototype ran on**). No roof-plane detail, but it still supports full inter-building shading and a flat-roof / fixed-tilt yield estimate. This mode is a first-class citizen, not an afterthought: it is both the honest fallback wherever LiDAR is unavailable (§13, India) and the direct continuation of the prototype's approach — proof the architecture degrades gracefully instead of breaking.

DESIGN PAYOFF

Framing LOD-1 as a *mode* rather than a limitation turns the prototype's constraint into an engineering feature: one provider-abstracted pipeline, two resolution regimes, an explicit and documented accuracy trade-off between them.

06 Data sources — US v1 (all public, all verified)

LAYER	RECOMMENDED SOURCE	ACCESS & NOTES
Building footprints	Microsoft Global ML Building Footprints (fallback: OSM via <code>osmnx</code>)	Bulk GeoJSON via GitHub <code>dataset-links.csv</code> , or STAC <code>ms-buildings</code> on Planetary Computer. License CDLA-Permissive-2.0. OSM is ODbL, coverage uneven.
Surface model (DSM)	USGS 3DEP LiDAR → derive DSM	Raw point cloud on AWS <code>usgs-lidar-public</code> (EPT/LAZ), read with PDAL <code>readers.ept</code> ; rasterize via <code>writers.gdal</code> , <code>output_type: max</code> . Shortcut: Planetary Computer hosts a pre-derived <code>3dep-lidar-dsm</code> STAC collection — use it where it covers your city.
Imagery	NAIP	Planetary Computer <code>naip</code> STAC, Cloud-Optimized GeoTIFF, 0.3–1 m, RGB+NIR, public domain. Revisit every 2–3 yrs (not for change detection).
Solar irradiance	NSRDB + PVWatts v8 (NREL, now NLR)	Free API key at <code>developer.nlr.gov/signup</code> . PVWatts converts tilt/azimuth/area → annual kWh. NSRDB irradiance at ~4 km (2 km for some GOES products).
Admin boundaries	US Census TIGER/Line	Free, no auth, <code>www2.census.gov/geo/tiger/</code> . Census tracts for aggregation; join via GeoPandas.

Naming note: the National Renewable Energy Laboratory was renamed the National Laboratory of the Rockies (NLR) on 1 Dec 2025; its developer API moved from `developer.nrel.gov` to `developer.nlr.gov` (old keys still valid). Use the new domain so the work reads current. Historical NREL reports keep their original technical-report numbers.

07 Technical pipeline

1. **Footprints** — pull building polygons for the study area (Microsoft Buildings or OSM).
2. **DSM from LiDAR** — PDAL pipeline: `readers.ept` → ground filter (`filters.smrf`) → `writers.gdal` `output_type:max` for the surface (buildings + canopy); `min` gives bare-earth DTM. Keep the DSM — every building at its real height — this is what makes inter-building shading possible (§7).
3. **Solar radiation on the surface** — compute per-cell insolation *with shadowing*. Three real options, no free lunch: (a) **GRASS** `r.sun` (open-source, scriptable from Python; horizon/shadowing via `r.horizon`) —

recommended for the raster surface; **(b) ArcGIS Raster Solar Radiation** (the "Esri way", Appendix); **(c) pvlib** for point-based per-plane irradiance. There is no pure-Python raster solar model — plan on GRASS for the surface pass and **pvlib** for the final per-plane estimate. Decide this up front.

4. **Roof-plane / orientation / tilt extraction** — clip points per footprint, then either RANSAC plane-fitting (classical baseline, same approach as Project Sunroof) or the ML route (§9). In block-model mode (§5) this step is skipped in favour of the extruded height.
5. **Filter usable roof area** — adopt a documented baseline rather than inventing thresholds: drop slope > 45°, radiation < 800 kWh/m²/yr, and north-facing surfaces (N. hemisphere); aggregate per building via zonal statistics. Because the radiation raster already carries shading, a heavily shaded roof falls below the threshold automatically.
6. **PV capacity & annual energy** — feed usable area + tilt + azimuth *and the shading-adjusted insolation from step 3* into PVWatts v8 → annual kWh, capacity factor; derive CO₂ offset from a grid-emissions factor. See §7 for why the shading value must pass through here, not raw location irradiance.
7. **Aggregate** — spatial-join per-building results into census tracts → choropleths of potential capacity, generation, and (with demographic joins) an equity view.

08 Inter-building & mutual shading — a first-class effect

In dense areas, shadows cast by *neighbouring* buildings are often the single largest correction to raw roof potential — bigger than tilt or orientation. **My SIH prototype already modelled this** (shadows cast across the Delhi block); this project preserves that rigour at city scale, but only because of one deliberate design choice worth stating explicitly.

THE MECHANISM

We model on a **DSM** — the full elevation surface, every building at its real height plus trees and terrain — not on isolated per-roof geometry. When the raster solar-radiation model runs over that DSM it performs **horizon / shadow-casting per cell**: for each point on a roof it tests whether a neighbour (or the roof's own parapet/dormer) blocks the sun along its path. Integrated across the day and year, this captures the time-varying reality — the tall building to the south shades your roof at 9 am in December but not at noon in June.

So height differences and adjacency *are* accounted for — via the DSM feeding steps 3→6. But this only works if the following are handled; each is a real trap:

1. **Use the DSM, never the DTM**, for the radiation pass. The bare-earth DTM has no buildings, so it zeroes out all inter-building shading.
2. **Buffer the study area**. A building on the AOI's south edge is shaded by a taller one just outside it. Pad the DSM by a margin sized to the tallest plausible shadow-caster, or edge buildings read as over-optimistic.
3. **Set an adequate shadow search distance**. Both `r.sun / r.horizon` and the ArcGIS tools cap how far they look for obstructions; downtown, a far tall tower still matters, so default radii can under-capture.

4. **Don't let PVWatts silently undo it.** PVWatts assumes an *unshaded* horizon. Feed it the shading-adjusted per-roof insolation (or convert to a shading-loss factor) — not the raw NSRDB location value, which discards the shading you just computed.
5. **Trees and seasonality are only partial.** A single leaf-on LiDAR/NAIP capture models summer tree shadows but not deciduous leaf-off in winter — a known, documentable bias.
6. **Resolution sets the ceiling.** At ~1 m LiDAR DSM you resolve building edges and genuine inter-building shadows. A coarse (e.g. 30 m) DEM cannot — one building shading its neighbour is sub-pixel. This directly bounds the block-model / India path (§5, §13).

Interview value: "here's how I handle mutual shading in dense blocks, and here's the subtle trap where PVWatts quietly discards it" is exactly the second-order reasoning that separates a modeller from a tutorial-follower.

09 Where the ML actually lives

Two legitimate options — do the first for depth, or start with the second and add the first as a stretch:

- **Deep-learning roof-plane & obstruction segmentation.** Train / fine-tune on **RoofN3D** (TU Berlin, free LiDAR roof-plane dataset); reference modern architectures (RoofSeg, SPPSFormer for point-cloud roof-plane instance segmentation; U-Net / Mask R-CNN for imagery). Segmenting obstructions (chimneys, vents, HVAC) that eat usable area is where accuracy — and an architectural eye — pays off.
- **Learned suitability / ranking model.** Gradient boosting or a small NN over engineered features (slope, aspect, irradiance, usable area, self-shading), trained against PVWatts- or Sunroof-style labels. Simpler to ship, still a defensible ML story, and pairs naturally with SHAP-style feature attribution for interpretability.

10 Vertical-facade BIPV STRETCH GOAL

The SIH prototype scored solar potential on *vertical facades*, not just rooftops — the "BI" in BIPV. Rooftop-only would silently drop that ambition, so it stays on the map as an explicit, clearly-scoped stretch rather than a v1 promise.

The honest engineering note is what makes this interesting: **LiDAR under-samples vertical surfaces** (it is a broadly top-down capture), so the high-resolution path is actually the *weaker* route for facades. The **block-model path (§5) is the more tractable one** — extruded LOD-1 walls give clean vertical surfaces, and the same DSM radiation pass already computes irradiance on them. This is a neat inversion: the prototype-heritage mode is the natural home for the prototype's most distinctive feature. Scope: facades as a Phase-2+ add-on on the block-model path, with per-facade irradiance from the raster surface and a documented accuracy caveat versus rooftops.

11 Delivery — stack-neutral

- **Publish results:** turn the GeoDataFrame into hosted layers (vector tiles / a hosted feature service) consumable by any web-map client.
- **Interactive deliverable:** a web map wrapped in a lightweight app — the artifact a reviewer actually clicks. Keep it framework-agnostic; the prototype's own web front-end (React/Node) is one option.
- **Esri route:** the full ArcGIS Online / Experience Builder / `arcgis.learn` path is a first-class supported deployment — detailed in the appendix so it doesn't couple the core pipeline to one vendor.

12 Phased roadmap

PHASE 0 · SCOPING & DE-RISKING — A FEW EVENINGS

Pick the study area (recommended: **Washington, DC** — dense open LiDAR, and a well-documented reference workflow exists for a DC neighbourhood to benchmark against). Confirm 3DEP coverage/vintage, get the NLR API key, stand up the repo and a data-access smoke test per source. Decide GRASS-vs-ArcGIS for the radiation step now.

PHASE 1 · MVP — ONE NEIGHBOURHOOD, END-TO-END · 2-3 WEEKENDS

One DC neighbourhood. Footprints + DSM + radiation (with shading) + RANSAC roof planes + PVWatts → a per-building suitability score and a clean static map. **Explicit goal:** reproduce and sanity-check against a published reference. Ship it even if rough — it proves the whole spine works.

PHASE 2 · FULL — CITY-SCALE + ML + DEPLOYED APP · MULTI-WEEK

Scale to the full District: tiling/batching for the whole point cloud, the ML segmentation model (RoofN3D-trained) for planes + obstructions, census-tract aggregation, an equity overlay, and the deployed web app. This is the portfolio centrepiece. Facade BIPV (§10) enters here as a stretch.

PHASE 3 · INDIA — RE-VERSION VIA MODULAR ADAPTERS · STRETCH

Swap data adapters behind the common interface for an Indian city (§13). Deliberately showcases modular, provider-abstracted design — and the block-model path (§5) that traces straight back to the SIH prototype.

13 India re-version — modular, with an honest boundary

Build a provider interface (`FootprintSource` , `ElevationSource` , `ImagerySource` , `IrradianceSource`); the core pipeline stays fixed, adapters change. The honest version of modularity is more impressive than pretending it's a clean swap.

LAYER	INDIA EQUIVALENT	SWAP QUALITY
Footprints	Google Open Buildings / Microsoft ML Footprints	Clean swap — as good or better than US
Irradiance	NASA POWER + PVGIS, with <code>pvl</code> replacing PVWatts	Clean swap (PVWatts' NSRDB strength is US-centric)
Admin boundaries	Indian census / admin polygons	Clean swap
Elevation / DSM	Copernicus DEM GLO-30 (30 m) / CartoDEM (~30 m)	Constrained — no open LiDAR; 30 m can't resolve roof planes
Imagery	Sentinel-2 (10 m)	Constrained — ~10–15× coarser than NAIP; a roof is a few pixels

THE HONEST BOUNDARY — SAY IT OUT LOUD

Footprints, irradiance, and aggregation are true adapter swaps. But DSM and imagery are limited by *resolution*, not just source — so the India version drops to the **block-model path (\$5)**: LOD-1 extruded footprints with a flat-roof / fixed-tilt heuristic, exactly the prototype's original approach. **Inter-building shading (\$7) is the sharpest casualty**: at 30 m a building shading its neighbour is sub-pixel, so mutual shading drops to a coarse sky-view-factor or horizon estimate — state the accuracy hit plainly. Naming this trade-off is the senior move; hiding it is the junior one.

14 Honest risks & caveats

1. **DSM isn't off-the-shelf for all of the US** — deriving it from raw 3DEP point clouds is real engineering (compute + volume). Pick a city with confirmed dense, recent LiDAR; lean on Planetary Computer's pre-derived DSM where it exists.
2. **No pure-Python raster solar model** — you take a GRASS (install/scripting overhead) or ArcGIS (licensing) dependency for the surface pass. Decide early.
3. **Shading is only as good as the DSM + settings** — DTM-instead-of-DSM, an unbuffered AOI, too-small a shadow radius, or passing raw irradiance to PVWatts each silently discards inter-building shading (\$7). Easiest mistakes to make, hardest to notice.

4. **Roof-plane extraction is the weakest link** — its errors propagate straight into capacity/kWh/CO₂. No large public ground-truth for arbitrary US cities (RoofN3D is NYC-specific), so validating a new city is nontrivial. Report uncertainty; don't over-claim precision.
5. **Cloud/free-tier limits are real** — hosted-layer publishing is often credit- or quota-metered; batch publishes and confirm caps before committing a platform as the final target.
6. **India ≠ "just swap the adapter"** — the geometry/imagery/shading stages need a functional downgrade to the block-model path, not a repoint. Scope it as such.

15 Reference benchmarks — compare & cite, don't reinvent

- **Google Project Sunroof** — the closest end-to-end precedent (CNN roof scoring → RANSAC planes → TMY shading → panel placement). Your "how mine differs" section.
- **NREL/NLR rooftop technical potential** — Gagnon, Margolis, Melius, Phillips & Elmore (2016), *NREL/TP-6A20-65298*: 1,118 GW / 1,432 TWh-yr, ≈39% of US electricity. Your quantitative sanity check.
- **DeepSolar** — Yu, Wang, Majumdar & Rajagopal, *Joule* (2018) — the flagship "CV + solar + city-scale" precedent (detects installed panels across 366k+ images).

APPENDIX A

e Esri deployment track

The pipeline above is vendor-neutral. Esri is a first-class supported deployment target and, for a GeoAI role, the most direct signal of fit — collected here so it stays decoupled from the core.

- **Publish results:** turn the GeoDataFrame into a Spatially Enabled DataFrame and push a hosted feature layer — `sdf.spatial.to_featurelayer(title=..., gis=gis)` (supports `overwrite=True` for refreshes).
- **Interactive deliverable:** ArcGIS Online Web Map → wrap in an **Instant App** (no-code) or **Experience Builder** (low-code).
- **"Esri way" modelling:** do the radiation with **Raster Solar Radiation** (upward hemispherical viewshed per cell, so surrounding buildings occlude the sky — note the legacy Area Solar Radiation tool is being superseded) and the segmentation with `arcgis.learn` (15+ built-in DL models on PyTorch/fastai) inside ArcGIS Pro — end-to-end ecosystem fluency.
- **Benchmark:** Esri's "Estimate solar power potential" tutorial (Glover Park, DC) — DSM + footprints → Raster Solar Radiation → slope/aspect filter → zonal stats — is the Phase-1 template and reference; plus Esri's public "Rooftop Solar Panel Suitability" Experience Builder app as a deliverable model.
- **Free-tier caution:** ArcGIS Online Public/Personal accounts are credit-metered per republish (2 GB storage). A city-scale layer fits, but repeated re-publishing burns credits — batch publishes; check `developers.arcgis.com/faq` before committing.

Esri-*application*-specific positioning (the interview narrative and the tailored resume copy) lives in the separate Esri project spec, not here — this document is the vendor-neutral engineering plan for the general portfolio and repo.

All technical claims, datasets, APIs, and benchmark figures were verified against primary sources (Aug 2026).

Companion: [SIH2024_BIPV_Solar_Prototype_Deck.pdf](#) (the prototype this project grows from). Sibling document: the Esri-tailored project spec.